- Thermal conductivity of elastomeric foam shall not exceed 0.035 W/ (m*K) at an average mean temperature of 24° C (Tested acc. To ASTM C177)
- Material shall be **anti-fungal** and shall pass resistance to Fungi test in accordance with **UL181.**
- Material shall be **anti-Bacterial** and shall pass bacterial resistance test in accordance with **ASTM E2180-01.**
- Material shall have **fire class A (Flame spread: 20 & Smoke developed index 150)** in accordance with **ASTM E 84**. It shall pass **smoke and toxicity** in accordance with IMO res. MSC 61(67) ann.
- It shall pass **rate of burning** (horizontal position) test in accordance with part-2 ASTM D635-06.
- Material shall have **flammability class V-0** in accordance to **UL94**.
- Material shall be **non-corrosive** for copper and stainless steel in accordance with DIN 1988.
- Material shall be complying with **RoHS (Restriction of Hazardous Substances)** Directive 2002/95/EC amended up to 2011/65/EU.
- The material shall be Dust and Fiber Free
- Unlike other generic materials it should be self-extinguishing, does not drip fireballs and does not spread flames
- Moisture Diffusion Resistance Factor or **'μ' value shall be minimum 7,000** per EN 12086 without any outer facing

Comply

- Material shall be **CFC/HCFC free** and shall be tested in accordance with **USEPA 5021A/USEPA8260C**
- Density of material shall be 40-60 Kg/m³ in accordance to ASTM-D-1667.
- Material should have service temperature range -50°C to +125°C.
- **Material Color: Black**

- **Summary of Duct Insulation Thickness**

	Insulation Thickness(mm)	Material to be used
Supply Air Duct	22	Closed cell anti-fungal & anti-bacterial elastomeric foam made from EPDM rubber with Alu foil
Return Air Duct	15	Closed cell anti-fungal & anti-bacterial elastomeric foam made from EPDM rubber with Alu foil

2. Duct Acoustic Insulation:

EPA approved Open Cell Nitrile Rubber Material with Microbial Agent for Anti-fungal and Anti-bacterial Properties

- Acoustic Insulation material Properties
- Sound Absorption Coefficient; tested as per IS: 8225/ISO: 354/ASTM 423C, shall be minimum as per enclosed chart
- Recommended Thickness for Ducts
- Application Procedure

- **Acoustic Material Properties**

Material: Open cell nitrile rubber sheet
Density: 180-220 Kg/m³

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- **Noise Reduction Coefficient (NRC)** value of insulation material should not be less than **0.55** as per testing standards in accordance to **ISO: 354/ASTM 423C.**

- Material shall have **fire class 1 in accordance with BS 476 part 7.**

- **Material shall be EPA approved and shall be Anti-Fungal & Anti-bacterial in accordance with ASTM G21 & ASTM E 2180.**

Freq (Hz)	100	125	160	200	250	315	400	500	630
Absorption Coefficient	0.01	0.03	0.04	0.05	0.09	0.16	0.20	0.32	0.45

Comply

Freq (Hz)	800	1000	1250	1600	2000	2500	3150	4000	5000
Absorption Coefficient	0.63	0.82	0.93	0.95	0.92	0.89	0.84	0.82	0.83

- **Recommended Thickness for Ducts**

  For Duct sizing lesser than 600 mm X 600 mm: **15 mm thick**

- **Application Procedure:**

  - Clean the surface to be insulated

  - Providing & Fixing Open cell nitrile rubber sheet having thickness of 15 mm inside the duct with the help of adhesive.

3. Under-deck Thermal Insulation

- Insulation material technical specifications
- Application procedure

- **Insulation Material Technical Specification:** 20 mm thick **Rigid PIR boards with both side company bonded FSK**

- PIR insulation boards shall have **density** of **45 kg/m^3** with tolerance as per **IS: 11239 (Part 2).**
- Thermal conductivity **(k-value)** of insulation material shall not exceed **0.023 W/mK at 10˚C mean temperature** as per testing standards in accordance to **ASTM C518.**
- Thermal resistance **(R-value)** of insulation material shall not exceed **4.937 ˚F ft^2 hr/BTU at 10˚C mean temperature.**

Comply

- Insulation material shall have **dimensional stability** up to **0.17%** at 100˚C for 24 hours as per testing standards in accordance to **IS:11239 (Part 3) / IS:12436.**
- **Water Absorption** shall be **0.61%** as per testing standards in accordance to **ASTM D2842.**
- **Compressive strength** shall be minimum **115 kN/m^2** as per testing standards in accordance to **IS:11239 (Part 11) / IS:12436.**

In Horizontal burning, **extension of burning** shall be **6 mm** as per testing standards in accordance to **IS: 11239 (Part 12) / IS:12436.**

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- **Water vapour permeability** shall be **0.017 ng/Pa.s.m** as per testing standards in accordance to **ASTM E96.**
- Size: **1.2 m x 1.2 m**

- **Application Procedure:**

Comply

- Clean the surface to be insulated.
- Providing and fixing **20 mm thick PIR boards with both sides FSK (Foil - Scrim - Kraft) foil** with the help of adhesive, G.I. cross wire, screw- washers & complete as per guidelines given in application manual.

4. AHU Acoustic Insulation:

- Insulation material technical specification
- Application Procedure

- **Insulation material technical specification: Ready-made acoustic panels with snap fit plugs**

- It shall have an overall thickness of **30 ± 2 mm.**
- It shall have length and width of **300 ± 2 mm.**
- It shall have weight of **700 to 750 grams** per unit.
- It shall be filled with material with acoustic properties and it shall have an NRC value of **0.70-0.75** in accordance with **ISO 354/ASTM 423C.**
- It shall also have an **STC** (Sound Transmission Class) value of **18 ± 2 dB.**
- It shall have **snap fit plugs** on the behind facing for fast application.

Comply

Application Procedure:

- The surface shall be cleaned.
- Holes shall be made on the walls in accordance to the dimensions of the acoustic panels.
- The perforated faced PP Acoustic panels shall be directly fixed on the wall with the help of its own snap fit plugs inside the holes on the walls.

5. Drain Piping Insulation:

- Insulation Material Properties
- Application Procedure

Comply

FM approved Closed cell Class O Nitrile Rubber - 13 mm thick sleeve without bonded cloth (Only Plain Nitrile Rubber sleeve to be applied with the help of specified adhesive.)

ion Material Technical Specification – FM approved Closed cell Class O Nitrile Rubber

ion material shall be **Closed Cell Elastomeric Nitrile Rubber.**

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